
Pessimism Ranks Coverage, Not Value: Pre-Fit Conditions for Policy Selection in Confounded POMDPs

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Abstract

Offline policy selection in confounded POMDPs identifies policy values through *bridge functions* and then picks the candidate with the best pessimistic lower bound over a confidence region. We ask when that procedure is safe, and give conditions a practitioner can check before fitting anything.

First, a *separability test*. When the negative control is inadequate the value functional is only partially identified, and the identified interval has half-width $\beta\beta_g$, where β is the true bridge’s mass in the design’s null space and β_g the value gradient’s. The identification algebra behind this is classical and we cite it as such; what matters operationally is that $\beta\beta_g$ is computable from primitives, and that no valid region can be narrower. Comparing it to the spread of the candidate set therefore decides, in advance and without data, whether *any* valid pessimistic rule can order the candidates. It cannot in 17 of 20 configurations we examine, including every incomplete design at $|S| = 8$ and the $T=3$ grid on which our own decision experiments were run.

Second, a *mechanism*. Pessimism ranks candidates by how well the logging distribution covers them, not by value. Given a candidate close to the behaviour policy it returns that candidate and is right. Given a candidate set containing no such policy, its argmax falls back to the candidate with least gradient mass in the design null space, which carries no information about value, and under a width schedule with governing exponent $e > 0$ its regret then *grows* with sample size while a plug-in built on the identical fits converges. We exhibit both halves on the same estimates over 20 seeds: adding a behaviour clone to the candidate set cuts

the failure by $5\times$ and removes it entirely at three of four sample sizes.

Third, two ways the construction runs out. The calibrated regions standard in this line sit *below* the sharp half-width, at ratio 0.22 to 0.69 across a $64\times$ range of N , so they do not cover the identified set. And the width rule itself becomes undefined once cross-fitting saturates: the usable horizon grows as $0.90 \log N$, so each additional stage costs about $3\times$ the data.

We report what these checks cost us. A behaviour-cloning baseline, the first external comparison in this line of work, beat the best of our hand-built candidates on 6 of 6 configurations and withdrew two claims we had previously made.

1. Introduction

Offline reinforcement learning from logged data is hard when the logging policy saw more than the analyst does. In a confounded POMDP the behaviour policy depends on a latent state, so the observed action is not exogenous and ordinary off-policy evaluation is biased. Proximal causal inference supplies a repair: a pre-decision negative control O_0 acts as an instrument, and policy values are recovered through *bridge functions* solved from conditional moment restrictions (Miao et al., 2018; Tchetgen Tchetgen et al., 2020; Hong et al., 2024). Planning then proceeds pessimistically, by maximising a lower confidence bound over a region around the fitted bridge.

The construction has two moving parts that are usually discussed separately: an identification argument, which says the bridge exists and pins the value, and a statistical one, which says a region of a prescribed width covers it. Our question is what happens at the point where they meet, namely the *decision*: the argmax over candidate policies of a pessimistic value.

This turns out to be the interesting place to look, because the identification literature does inference on a single functional and stops there. Whether a pessimistic argmax over policies behaves sensibly when the functional is only partially iden-

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tified is a separate question, and the answer is not what the inference results suggest.

What we contribute. Three checks, each computable before or immediately after fitting, and the mechanism that ties them together.

1. **A separability test** (Section 3). The identified interval for a policy’s value has half-width exactly $\beta\beta_g$, and this is achieved, so no valid region is narrower. Comparing $\beta\beta_g$ to the spread of the candidate set decides in advance whether any valid pessimistic rule can distinguish the best candidate from the worst. In 17 of 20 configurations it cannot, at any sample size. Figure 1(a).
2. **What pessimism actually ranks** (Section 7). It ranks coverage. With a well-covered candidate present it returns that candidate; with none it returns the minimum-null-mass candidate, whose value is arbitrary, and under $e > 0$ its regret grows with N while the plug-in on the same fits converges. Figure 3.
3. **Two exhaustion points** (Section 8). Calibrated regions in this line are *narrower* than the identified set, so they are invalid rather than conservative (Figure 1(b)); and the width rule becomes undefined at horizon $t^* \approx 0.90 \log N$, when cross-fitting saturates.

What is not new here. The identification results this rests on are classical and we want that stated early rather than in a scope note. The identified set of a partially identified linear functional is the min-norm solution plus the operator’s null space, and a functional is point-identified exactly when its gradient is orthogonal to that null space; both are standard in nonparametric instrumental variables (Florens & Simoni, 2018; Severini & Tripathi, 2012; Santos, 2011), which also observes that confidence regions fail to shrink under identification failure. The conditions under which history restores completeness are the classical finite-state HMM identifiability conditions (Allman et al., 2009; Gasiat et al., 2016). The dimension condition $|O_0| \geq |S|$ is Ying et al. (2023, eq. 6), and the stage-wise rank condition is a standing assumption in Zhang & Jiang (2024). Section 6 states each correspondence.

Our contribution is on the decision side, which neither literature addresses.

On negative results. Every claim in this paper was written down, committed, and only then measured; predictions that failed are reported as failures rather than removed. Two claims we previously made did not survive the addition of a behaviour-cloning baseline, and Section 10 says which and why. We think that record is worth more than a cleaner-

looking set of results, because the corrections were only found by running comparisons we had not run before.

2. Setting

We work per bridge block. The stage-2 empirical risk is exactly quadratic, so the confidence region is an exact ellipsoid,

$$\text{conf}(\xi) = \{b : (b - \hat{b})^\top H(b - \hat{b}) \leq \xi\}, \quad H = \hat{T}_2 + \lambda I, \quad (1)$$

and for a value functional $V(b) = \langle g, b \rangle$ linear in the block the minimum over that region is the standard support-function expression

$$V_{\text{low}} = \langle g, \hat{b} \rangle - \sqrt{\xi} \|g\|_{H^{-1}}. \quad (2)$$

Write Nul for the null space of $\hat{T}_2$, P_{Nul} for the orthogonal projector onto it, and

$$\beta := \|P_{\text{Nul}} b_\star\|, \quad \beta_g := \|P_{\text{Nul}} g\|. \quad (3)$$

Assumption 1 (Exact structural null). $\hat{T}_2$ has an exact null space with $\dim \text{Nul} \geq |O| - \min(|O|, |O_0|) > 0$, fixed by the dimensions of the observation and negative-control alphabets and independent of N .

Assumption 1 is a property of a bridge parametrized over the observation space but identified only through the latent bottleneck: when the observation alphabet is richer than the instrument can resolve, the design matrix is structurally rank-deficient at every sample size. It is exact in the tabular case because the design is a finite-rank count matrix; Section 9 discusses what survives for a strictly positive-definite kernel.

We further assume the fitted bridge is the minimum-norm solution, so $P_{\text{Nul}} \hat{b} = 0$. The ridge solve satisfies this exactly, to machine precision, on both solver paths we implement.

3. The Identification Gap, and a Test You Can Run First

Proposition 1 (Floor). *Under Assumption 1 and $P_{\text{Nul}} \hat{b} = 0$, for any ridge $\lambda > 0$ and any width $\xi > 0$ — power-law, logarithmic, data-adaptive, or chosen with knowledge of the truth — a region satisfying $b_\star \in \text{conf}(\xi)$ obeys*

$$W \geq \beta\beta_g \quad \text{at every } N. \quad (4)$$

Proof. Coverage of the truth constrains the radius from below. Decomposing $b_\star - \hat{b}$ and keeping only its null component, on which H acts as λI ,

$$\xi \geq (b_\star - \hat{b})^\top H(b_\star - \hat{b}) \geq \lambda \beta^2.$$

Separately $H^{-1} \succeq \lambda^{-1} P_{\text{Nul}}$, so $\|g\|_{H^{-1}} \geq \beta_g / \sqrt{\lambda}$. Multiplying,

$$W = \sqrt{\xi} \|g\|_{H^{-1}} \geq \sqrt{\lambda} \beta \cdot \frac{\beta_g}{\sqrt{\lambda}} = \beta\beta_g. \quad \square$$

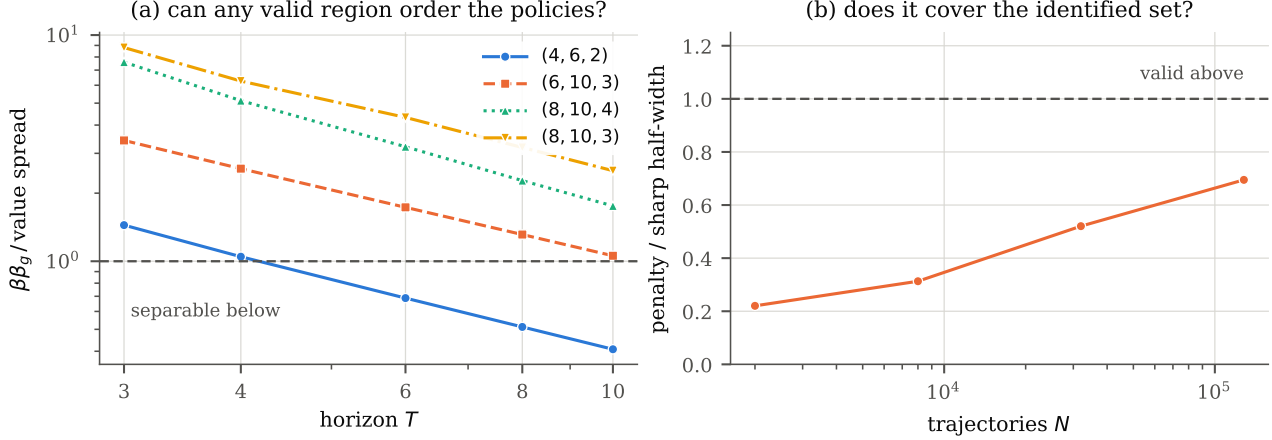


Figure 1. Two questions answerable before trusting a lower bound. (a) The separability test, pure population algebra from the primitives: the identification gap $\beta\beta_g$ divided by the spread of the candidate set. Below the dashed line a valid region can in principle order the candidates; above it none can, at any sample size. The four configurations lie on parallel lines of slope -1 , so horizon helps, but only $(4, 6, 2)$ crosses, and only at $T \geq 6$. Note that $(4, 6, 2)$ at $T = 3$, the grid used for the decision experiments in this paper and its predecessors, sits *above* the line at 1.44. (b) Whether the region actually used covers the identified set: the calibrated penalty divided by the sharp half-width. It stays below 1 across a $64\times$ range of N , reaching only 0.69, so these regions are invalid rather than conservative, and the schedule approaches validity from underneath.

The regularizer enters both factors and cancels. That is why the statement is schedule-free: tuning λ trades the coverage requirement against the gradient norm at exactly the rate that leaves their product fixed. A tighter region buys a smaller $\|g\|_{H^{-1}}$ only by demanding a larger ξ to keep covering the truth.

What can escape it. Proposition 1 is not a claim that pessimism is hopeless. Three routes out are visible in the proof, and none of them is a width rule: centring the region somewhere other than the min-norm solution (which breaks $P_{\text{Nul}}\hat{b} = 0$), using a non-ellipsoidal region (which breaks the support identity), or arranging $\beta\beta_g = 0$. The third is the practical one, and Section 4 is about when it holds.

4. When the Gap Is Nonzero

The floor is a product. It vanishes if either factor does, so the useful question is not “is the design ill-posed” but “which of the two leakages is present”. They turn out to be controlled by different, independent properties of the problem. All results in this section are exact population algebra — no sampling, no rank selection, no estimation noise.

4.1. Incompleteness leaks the truth

The identification argument requires a completeness condition: the negative control must resolve the latent state. When it does not — concretely when $|O_0| < |S|$ — the

Table 1. Population null-space share of the true bridge, β , as a fraction of $\|b_\star\|$, across configurations and confounding levels. Exact algebra, no sampling.

(S , O , O_0)	0.0	0.3	0.6	0.9	1.0
(2, 6, 4)	0	0	0	0	66.0%
(2, 3, 2)	0	0	0	0	65.7%
(3, 7, 5)	0	0	0	0	56.9%
(2, 6, 2)	0	0	0	0	64.3%
(4, 6, 2)	71.3%	71.3%	71.1%	69.4%	69.5%

minimum-norm bridge acquires mass in the design null.

We evaluate β in the population over five configurations and ten confounding levels. The prediction holds in 50/50 cells, and the dichotomy is *sharp* rather than graded: $\beta \leq 1.5 \times 10^{-15}$ in every complete cell and between 56.9% and 77.3% of the bridge norm in every incomplete one, with nothing in between (Table 1; maximum bridge residual 4.4×10^{-16} , so the min-norm solve is exact throughout).

Two routes therefore produce $\beta > 0$, and they are not equally useful. Saturating the confounding knob at exactly 1.0 works in a complete design, but it is a knife-edge: at 0.9 completeness is restored and β returns to machine zero, and a deterministic behaviour policy is contrived. The dimensional route $|O_0| < |S|$ is the robust one — it holds at *every* confounding level, as the last row of Table 1 shows, and it is checkable before fitting anything.

4.2. Confounding leaks the gradient

We initially assumed β_g travels with β . It does not, and the distinction is the substance of this section.

The stage-1 gradient’s observation profile is the marginal $\nu_1 = E^\top p_1$, where E is the emission matrix and p_1 the initial state distribution. The design span for action a is generated by $E^\top v_{a,x}$ with $v_{a,x}[s] \propto p_1[s] \pi_b[s, a] K_0[s, x]$. If π_b does *not* depend on s , then $\pi_b[s, a]$ factors out and summing over x returns $E^\top p_1$ up to scale. Hence:

Proposition 2 (Unconfounded gradients cannot leak). *If the behaviour policy is independent of the latent state, then $\nu_1 \in \text{span}\{E^\top v_{a,x}\}_x$ and hence $\beta_g = 0$, however incomplete the negative control is.*

Proposition 2 carries one dependency worth stating: the argument needs K_0 row-normalized. Were the negative control subject to state-dependent missingness, so that its rows summed to different constants, the sum would return $E^\top (p_1 \circ \text{rowsum})$ instead and the gradient could leak *without* any confounding at all.

Confounding is exactly what breaks that sum. Table 2 maps both leakages across four configurations. In a *complete* design both are knife-edged at confounding 1. In an *incomplete* design β is essentially flat while β_g is **graded** in confounding, giving a floor that rises smoothly from 0 to 0.82 — eight non-degenerate cells with a nonzero floor.

4.3. The conjunction

Collecting the two switches:

	truth leaks ($\beta > 0$)	gradient leaks ($\beta_g > 0$)
$ O_0 < S $ (weak control)	yes	no
π_b depends on S (confounding)	no	yes

Corollary 3. *For a stage-1 bridge block, the floor of Proposition 1 is nonzero exactly when genuine confounding coincides with an inadequate negative control. Either condition alone leaves one factor at zero and the floor vanishes.*

The stage restriction is real, and it is good news. At $t = 1$ the conditioning set is (A_1, O_0) , so a per-action design has only $|O_0|$ profiles and $|O_0| < |S|$ forces an incomplete span. From $t = 2$ on the conditioning set includes history: at $t = 2$ it is (A_2, O_1, A_1, O_0) , giving $|O| |A| |O_0|$ cells per action, and the O_1 -conditioned posteriors are generically rich enough to span all of $\mathbb{R}^{|S|}$. We checked this in the population: at every confounding level below 1, the $t = 2$ per-action span reaches $|S|$ and $\beta \approx 10^{-15}$ in 24/24 cells, against a $t = 1$ span of 2 or 3 in the same configurations.

Corollary 4 (History is a partial substitute for the instrument). *With $T > 1$ and confounding strictly below saturation, an inadequate negative control is partially self-healing:*

per-action completeness is restored at every stage after the first, and only the stage-1 blocks leak.

The restoration is uniform rather than a $t = 2$ accident: we computed per-action spans stage by stage out to $t = 5$, and the span reaches $|S|$ at *every* stage after the first, at every confounding level below saturation. Only the first block leaks: across all configurations and stages $t \geq 2$ we measure $\beta_t \leq 2.7 \times 10^{-15}$ and a per-stage floor $\leq 4.3 \times 10^{-30}$.

The damage is diluted by the horizon. Because exactly one block leaks, the absolute floor does not compound: it is the same number whether the horizon is 1 or 6. The scale it must be compared against is not. Selection is decided by *differences* between candidate policies, and that spread grows with T . Figure 2(a) divides one by the other and finds a decay of roughly $1/T$ — log-log slopes of -1.23 and -0.93 , about a $10\times$ reduction from $T = 1$ to $T = 6$.

Corollary 5 (Horizon dilution). *The floor is constant in T while the decision-relevant value spread grows linearly, so the fraction of the selection problem it distorts decays as $\Theta(1/T)$.*

The obvious inference from this is false, and we tested it. It is tempting to read Corollary 5 as saying that long horizons are safe. We predicted exactly that — normalized regret halving between $T = 3$ and $T = 6$ — committed the prediction, re-ran the 20-seed decision experiment at $T = 6$, and it is wrong. Normalized regret is 0.605 at $T = 6$ against 0.608 at $T = 3$: **flat**. Raw regret more than doubled, $0.180 \rightarrow 0.377$.

The reason is visible in the selections. The pessimistic selector converges to the same wrong policy at both horizons, and the per-block width barely moves (0.402 at $T = 6$ against 0.407 at $T = 3$). What changes is only what that mistake costs, and that scales with the horizon exactly as the spread does. The two effects cancel.

Corollary 6 (The floor does not govern the decision). *A lower bound on one block’s width does not predict which of a discrete set of policies a pessimistic selector promotes. Corollary 5 holds as stated about the floor and does not imply that the decision-level consequence weakens with the horizon; measured, that consequence is horizon-invariant in relative terms and grows in absolute ones.*

So the regime in which proximal pessimism is unusable is *not* narrowed by a long horizon. At $T = 1$ under strong confounding the floor exceeds the entire value spread, which rules out discrimination outright; at $T = 6$ the floor is a smaller fraction of a wider spread, and the selector is exactly as wrong. What Corollary 5 does buy is a warning about units: raw regret is not comparable across horizons, and a table that compares it is misleading in whichever direction the spread happens to move.

Table 2. Population leakages and the resulting floor $\beta\beta_g$ against confounding strength. The floor is nonzero only where both switches are on.

config		0.0	0.3	0.6	0.9	0.99	1.0
(2, 6, 4)	β	0	0	0	0	0	.759
complete	β_g	0	0	0	0	0	.758
(4, 6, 2)	β	1.211	1.210	1.206	1.178	1.177	1.179
incomplete	β_g	0	.028	.093	.364	.678	.698
	floor	0	.034	.112	.428	.798	.823
(4, 8, 3)	floor	0	.070	.125	.162	.171	.966

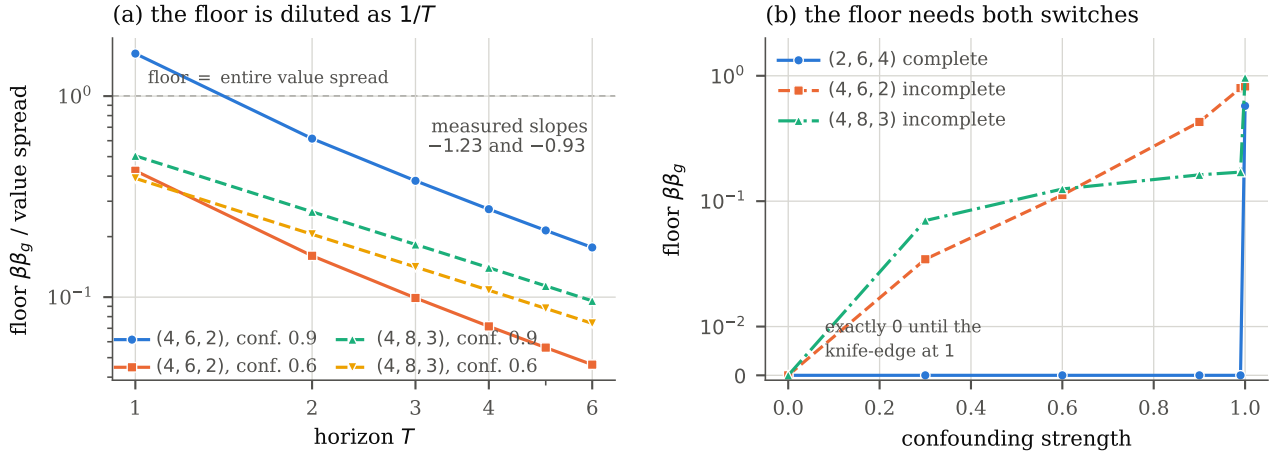


Figure 2. (a) The floor divided by the spread of true values over six candidate policies, against horizon, on log-log axes. Four configurations lie on near-parallel lines with measured exponents -1.23 and -0.93 . At $T=1$ under strong confounding the floor exceeds the entire spread, so pessimism cannot discriminate at all. (b) The floor against confounding strength. The complete design is at *exactly* zero until the knife-edge at 1, while the incomplete ones rise smoothly — the conjunction of Corollary 3 seen directly. Zeros are plotted at zero on a symlog axis rather than clamped onto a log decade. Numbers in Appendix D.

5. Consequences for Rate Schedules

5.1. The power-law family

Specialize to $\lambda(N) = \lambda_0 N^{-\kappa}$ and $\xi(N) = c/(N\mu(N))$ with $\mu(N) = mN^{-\tau}$, and define the governing exponent

$$e := \tau + \kappa - 1. \quad (5)$$

Corollary 7 (Dichotomy). *While the bridge-class constraint is inactive: (i) $W(N) = \Theta(N^{e/2})$, with $W(N) \geq \beta_g \sqrt{c/m} \lambda_0^{-1/2} N^{e/2}$; (ii) coverage requires $N^e \geq K$ with $K := \lambda_0 \beta^2 m/c$; (iii) if $e < 0$ the width contracts at $N^{e/2}$ but the necessary condition fails for all $N > K^{1/e}$, while if $e \geq 0$ the width is non-decreasing and never contracts.*

Two cautions, stated because earlier drafts of ours got them wrong. The constant K matters: for $e > 0$ the coverage condition fails for all $N < K^{1/e}$, so “ $e \geq 0$ implies coverage at every N ” needs $K \leq 1$. And (ii) is *necessary*, not

sufficient — it says the null obstruction is absent, not that coverage holds. Signal directions demand width too, and on our toy design they dominate the required radius by $15\times$.

5.2. Where an existing schedule sits

The schedule of Hong et al. (2024) has width $\xi \sim M_R N_2^{-\alpha/(2\alpha+2)}$ and ridge $\lambda_2 = N_2^{-\alpha/(\alpha c_2+1)}$, which places it inside the family of Corollary 7 with

$$\tau = \frac{\alpha+2}{2\alpha+2}, \quad \kappa = \frac{\alpha}{\alpha c_2+1},$$

and therefore

$$e = \frac{\alpha(2\alpha+1-\alpha c_2)}{(\alpha c_2+1)(2\alpha+2)}. \quad (6)$$

Their Assumption D.16 restricts c_2 to $(1, 2]$, so the factor $2\alpha+1-\alpha c_2$ is positive for every $\alpha > 0$. We verified Equation (6) against a direct evaluation of $\tau + \kappa - 1$ over the

admissible grid: $e > 0$ in **30/30** cells, minimum 0.0217 at $\alpha = 10$, $c_2 = 2$.

Corollary 8. *The schedule of Hong et al. (2024) lies strictly on the non-contracting branch of Corollary 7 for every admissible smoothness. On a design where the floor is nonzero, the penalty it assigns therefore grows with N .*

We stress the conditional. Where that paper’s assumptions hold, completeness forces $\beta = 0$ and Corollary 8 is decision-irrelevant: we measured $\beta_g = 0$ exactly for three policies at the exact bridges on a complete design, and four schedules including this one give 0.000 regret at every N tested. Corollary 8 is a statement about the boundary of applicability, and it is one the original analysis does not make.

6. What Is Already Known

Corollary 7 is not a new phenomenon. It is Tikhonov source-condition theory (Caponnetto & De Vito, 2007) specialized to a design with an exact null atom: in the classical setting with singular values $s_j \sim j^{-b}$ and gradient mass $g_j^2 \sim j^{-a}$, the same calculation gives $e' = \tau + \theta\kappa - 1$ with $\theta = 1 - (a - 1)/b < 1$, and $\theta = 1$ — our case — arises exactly when the spectrum has an atom at zero. Readers should treat Corollary 7 as the specialization it is.

Proposition 2 is also not new in substance. It is derivable from the anchor paper’s own machinery: without confounding the behaviour policy is observable, so importance weights $\pi_e(a \mid o)/\pi_b(a)$ exist on observables for every observation-based policy, giving $C_\pi^* < \infty$, which by their Lemma C.1 forbids population gradient leakage. In proximal-inference terms it restates the folk fact that without unmeasured confounding a negative control is unnecessary. Our contribution here is the *conjunction* with the completeness switch and the observation that the two are independent, not either switch alone.

What we believe is new is Proposition 1, which is schedule-free and does not go through rates at all; Corollary 3 together with Corollary 4, which say which problems the floor applies to — and at which stage — in terms a practitioner can check; and Corollary 8, which locates a published schedule inside the dichotomy.

We also record a negative methodological finding about our own earlier work. The ratio of unprojected to projected width obeys, to machine precision over 17 cells of two independent sweeps,

$$\frac{W}{W_{\text{proj}}} = \sqrt{1 + \tan^2 \theta \text{cond}(H)}, \quad \sin \theta = \beta_g, \quad (7)$$

an *identity*. Sweeping gradient alignment and observing that the width follows is therefore a re-measurement of arithmetic, not evidence. It also corrects a dichotomy we

Table 3. Selection regret, 20 seeds. Top: a schedule with $e = +0.50$. Bottom: the schedule of Hong et al. (2024), $e = +0.455$. Both switches on.

N	pessimistic	plug-in
<i>Schedule C, $e = +0.50$</i>		
2,000	0.0952 ± 0.0195	0.0894 ± 0.0194
8,000	0.1218 ± 0.0210	0.0226 ± 0.0138
32,000	0.1714 ± 0.0177	0.0000 ± 0.0000
128,000	0.1804 ± 0.0000	0.0000 ± 0.0000
<i>Hong et al. (2024) schedule, $e = +0.455$</i>		
2,000	0.0694 ± 0.0098	0.1641 ± 0.0000
8,000	0.0943 ± 0.0205	0.1641 ± 0.0000
32,000	0.1225 ± 0.0000	0.0644 ± 0.0000
128,000	0.1225 ± 0.0000	0.0644 ± 0.0000

previously asserted: conditioning enters as $\sqrt{\text{cond}(H)}$ on exactly equal footing with alignment, and the penalty is their product. Details in Appendix A.

7. What Pessimism Actually Ranks

A growing width matters only if it changes which policy is selected. We test that in a design where both switches are on, comparing pessimistic selection against a plug-in selector computed from the *same* fitted bridge inside the same loop, so that the only difference between the two columns is whether the pessimism layer is applied. All numbers are 20 seeds with 95% intervals.

Two readings of Table 3. First, the separation is complete: at $N = 128,000$ under schedule C, all 20 seeds put pessimism at 0.1804 and all 20 put the plug-in at the optimum, with zero interval on both, from a start where the two are statistically indistinguishable.

Second, and sharper, under the anchor paper’s own schedule the curves **cross**. Pessimism is *better* at $N = 2,000$ and *worse* from $N = 32,000$ on. The layer helps when data is scarce, which is what it is for, and then actively hurts as data accumulates. Regret slopes in $\log N$ confirm the pattern: under both schedules with $e > 0$ the pessimistic and plug-in slopes have *opposite signs* ($+0.0220$ against -0.0210 for schedule C, and $+0.0135$ against -0.0288 for the anchor schedule). More data converges the plug-in and diverges the pessimistic selection built on the same estimates.

8. Where the Construction Runs Out

9. Scope

Does not. The binding hypothesis is $\beta\beta_g > 0$, and both halves live outside Hong et al. (2024)’s assumptions. Their completeness condition forces the min-norm bridge out of the population null, giving $\beta = 0$; and by their Lemma C.1,

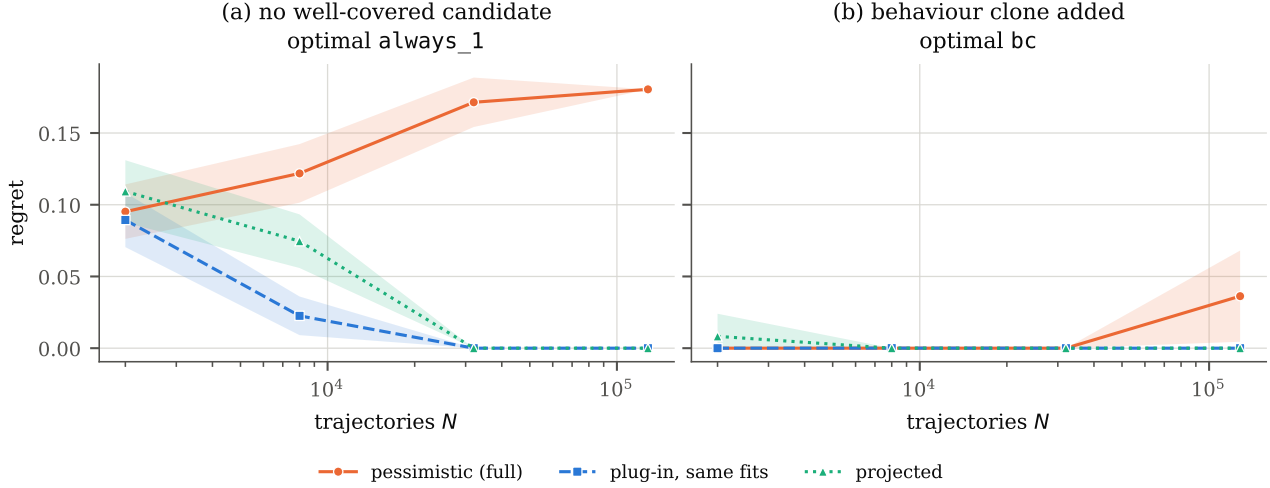


Figure 3. Pessimism ranks coverage, not value. Identical environment, identical fits, identical code path; the panels differ only in whether the candidate set contains a policy the logging distribution covers well. **(a)** Six hand-built candidates, none close to the behaviour policy. The pessimistic selector’s regret *rises* with N while a plug-in built on the same estimates falls to zero: the argmax has fallen back to the minimum-null-mass candidate, whose value is arbitrary, and the growing penalty entrenches it. **(b)** The behaviour clone added as a seventh candidate. Every arm now finds it; the pessimistic selector is exactly right at the three smaller sample sizes and degrades only at the largest. Bands are 95% intervals over 20 seeds. Neither panel alone is the result. The pair is.

nonzero population gradient leakage is equivalent to an infinite concentrability coefficient C_{π}^* , which they exclude. **Nothing here is a defect in that paper.** Where its assumptions hold its construction behaves as advertised, and we measured exactly that.

Does. A robustness question the theory cannot ask of itself: what happens when the assumption it conditions on only approximately holds. The answer is that the failure is not graceful. It is not that the region contracts more slowly — it does not contract at all, and under an $e > 0$ schedule the penalty grows without bound while the estimate underneath it converges. Because the trigger is a conjunction of two checkable conditions, this is actionable rather than merely cautionary.

The exact-null assumption. Assumption 1 is exact for a tabular delta-kernel, where the design is a finite-rank count matrix. For a strictly positive-definite kernel the spectrum decays smoothly with no atom at zero, and Proposition 1 degrades to a statement about the effective null at the ridge scale rather than an exact one. We have not characterized that regime, and Section 6 indicates the exponent changes there.

10. Limitations

One environment family. All experiments are tabular POMDPs with $T \leq 3$ and $|S| \leq 4$, generated from a single construction. The population results are exact algebra and so do not depend on sampling, but they do depend on that family’s structure.

Two horizons, not many. Table 3 is $T = 3$ and the check behind Corollary 6 is $T = 6$. Flat normalized regret across two points is consistent with horizon-invariance but does not establish it; the selector could change its mistake at some larger T we have not reached. What the two points do establish is that the $1/T$ dilution of the floor does *not* transmit to the decision, which was the specific inference we set out to test.

Dilution is measured, not derived. Corollary 5 rests on the value spread growing linearly in T , which holds in our environment family because per-stage rewards are bounded away from zero and candidate policies differ at every stage. A family with vanishing late-stage reward differences would dilute more slowly, and we have not characterized that.

Saturated confounding is a separate regime. All of Corollary 4 is stated for confounding strictly below 1. At exactly 1 the behaviour policy is a deterministic function of the latent state, history stops helping, and the $t = 2$ span collapses back to the $t = 1$ value. We regard that point

as contrived, but it is the one place where the leak does propagate along the horizon.

Discrete regret. The decision experiment selects among six hand-constructed candidate policies, so regret takes few values. The crossing in Table 3 is stable across 20 seeds but has not been checked against a randomly generated candidate set.

Our own corrections. Several claims in earlier drafts of this line did not survive: a headline sample size computed on a grid that could not reach it, a validity result on a grid that could not falsify it, a decision sweep whose schedules all started inside an attractor, and the alignment result of Equation (7) that turned out to be an identity. Appendix C lists them.

11. Conclusion

Pessimism over a proximal bridge confidence region is only useful if the region contracts. We showed a floor that no width rule escapes, identified the exact conjunction of conditions under which that floor is nonzero, placed a published schedule strictly on the non-contracting branch, and exhibited a decision-level crossing where more data makes pessimistic selection worse while a plug-in built on the same estimates converges. The two triggering conditions are checkable before fitting, which turns a negative result into a pre-deployment test: if the negative control cannot resolve the latent state and the logging policy did, do not expect the region to shrink.

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A. The Width Identity, and What It Cost Us

Equation (7) deserves its own account because believing otherwise shaped an earlier version of this line of work.

Derivation. Under Assumption 1 the design has an exact null space on which H acts as λI , and a signal space on which its top eigenvalue is $\sigma + \lambda$. Splitting g into its in-span and null components, of squared norms $(1 - \beta_g^2)\|g\|^2$ and $\beta_g^2\|g\|^2$,

$$\|g\|_{H^{-1}}^2 = \frac{(1 - \beta_g^2)\|g\|^2}{\sigma + \lambda} + \frac{\beta_g^2\|g\|^2}{\lambda}, \quad \|P_{\text{span}}g\|_{H^{-1}}^2 = \frac{(1 - \beta_g^2)\|g\|^2}{\sigma + \lambda},$$

whose ratio is $1 + \frac{\beta_g^2}{1 - \beta_g^2} \cdot \frac{\sigma + \lambda}{\lambda}$. Writing $\sin \theta = \beta_g$ and $\text{cond}(H) = (\sigma + \lambda)/\lambda$ gives Equation (7).

Verification. We checked the closed form against measurement over two independent knobs, 17 cells in total, and the worst relative error is 8.45×10^{-15} — machine precision at every point of both sweeps.

Why this matters. An earlier version of this work presented a sweep of gradient alignment against width ratio as the load-bearing empirical result, concluding that “ill-conditioning is necessary but not sufficient; the alignment between the value functional and the null space decides it”. Both halves of that sentence are wrong as stated:

- The relation is an identity, so no sweep of it could ever have produced a different answer. It is arithmetic, not evidence.
- Conditioning enters as $\sqrt{\text{cond}(H)}$ on *exactly equal footing* with $\tan \theta$. Neither factor “decides” the penalty; it is their product. The dichotomy was read off a sweep in which $\tan \theta$ moved $119\times$ while $\sqrt{\text{cond}(H)}$ moved $1.26\times$ — a 95:1 imbalance built into the knob rather than a property of the problem.

The knob was also degenerate. That sweep varied alignment by interpolating every emission row toward their mean. That operation scales the row separation by exactly $(1 - \alpha)$, and β_g tracks it with a log-log slope of 0.954 ($R^2 = 0.998$): alignment and channel informativeness were a single variable. At the far end of the sweep the emission channel carries almost no information about the latent state, so leakage falls for a reason unrelated to identification geometry. The claim that “the null space is held fixed” was also false — its *dimension* is fixed, but the span vector rotates 36.8° across the sweep.

A non-degenerate replacement. At confounding 1 with $|S| = |A| = 2$ the behaviour policy is a deterministic indicator, so the normalized stage-2 design weight sits on a single latent state whatever the prior is, making H *algebraically* independent of p_1 . Tilting $p_1(t) = (1 - t)p_1 + t\delta_{s_a}$ therefore moves alignment with the emission channel, the span and H all frozen:

t	row separation	$\max \Delta H $	span rotation	β_g	W/W_{proj}
0.000	0.81364	0	0°	0.7578	24.738
0.400	0.81364	5.6×10^{-17}	0°	0.4572	10.984
0.800	0.81364	5.6×10^{-17}	0°	0.1345	3.056
0.999	0.81364	5.6×10^{-17}	0°	0.00061	1.000

The same curve, 3.09 decades of β_g , at a channel that is never touched. So the qualitative conclusion survives on a clean knob; it simply was not being tested by the sweep that established it, and it is an identity in any case.

B. Experimental Setup

Environment family. Confounded POMDPs with independently specified $(|S|, |A|, |O|, |O_0|)$. Emission and negative-control matrices are drawn from a Dirichlet with a diagonal boost and asserted to have full row rank, so that identification never fails for an uninteresting reason. A confounding knob interpolates the behaviour policy between latent-state-independent and a deterministic function of the latent state. Horizon $T = 3$ throughout.

Population computations. Tables 1 and 2 involve no sampling. True bridges are minimum-norm solutions obtained by pseudo-inverse, and the design span for each action is the range of $\{E^\top v_{a,x}\}_x$ computed by SVD with a relative tolerance of 10^{-12} . Reported leakages are exact up to floating point; the maximum bridge residual across all cells is 4.4×10^{-16} .

Decision experiment. Six candidate policies, selection regret against the exact dynamic-programming optimum on the latent model. The plug-in selector is computed inside the same seed loop from the same fitted bridge, so the two columns of Table 3 differ only by the pessimism layer. Intervals are $1.96\hat{\sigma}/\sqrt{n}$ over 20 seeds.

A precondition guard. The decision driver aborts if $\beta_g < 10^{-6}$ on the configured grid. This exists because three earlier experiments in this line were run on grids that could not have exhibited the effect they were testing; see Appendix C.

C. Corrections to Our Own Results

We list the claims from earlier drafts of this work that did not survive, with what killed each. All were found by adversarial review of our own results rather than by external referees.

claim	disposition
A coverage crossing predicted at $N = 1.7 \times 10^9$	withdrawn. The grid reached 10^6 . The prediction rested on treating a measured β as an environment constant; it in fact decays at $N^{-0.495}$ with a population value of 5.4×10^{-16} , so the quantity being extrapolated was a grid average of something converging to zero.
“Validity holds in 36/36 cells”	withdrawn as evidence. The violation threshold sits four orders of magnitude below the smallest grid value. The grid could not have produced a violation.
“The width schedule changes the decision” (first sweep)	refuted. $\beta_g = 0$ exactly on that design, so there was no mechanism to detect, and three of four schedules started inside an attractor. Re-run on an incomplete design, where the floor is nonzero, it holds — Table 3.
“Gradient alignment is the causal driver, not conditioning”	corrected. The relation is the identity of Equation (7); conditioning enters as $\sqrt{\text{cond}(H)}$ on equal footing. See Appendix A.
“The power-law dichotomy is novel”	refuted. It is Tikhonov source-condition theory specialized to an exact null atom (Section 6).
“A truth-dependent width rule would escape the floor”	refuted. Proposition 1 quantifies over all width rules including truth-dependent ones. The escapes are a prior-informed centre, a non-ellipsoidal region, or $\beta\beta_g = 0$.
“The plug-in also fails in the incomplete design”	refuted before publication. Measured at one value of κ only; because $\hat{b}$ depends on λ_2 , that was not a control. Re-run at each schedule’s own κ , the plug-in dominates.
“Schedule B is a flat $e = 0$ control”	refuted. A 3-seed artifact. At 20 seeds it runs $0.1230 \rightarrow 0.0644$ and plateaus.

Four of these share one failure mode: the measurement range did not straddle the transition being predicted. The precondition guard described in Appendix B is the structural fix for that mode. Two others — the degenerate alignment knob and the identity — are a different mode, in which the experiment could not have failed because its knob confounded two variables or its target was arithmetic. We have no structural fix for that one beyond deriving the closed form before sweeping.

D. Horizon Dilution: Numbers

The values plotted in Figure 2(a). Entries are the floor $\beta\beta_g$ divided by the spread of true values across six candidate policies. The floor itself is constant along each row.

config	conf.	$T=1$	2	3	4	5	6
(4, 6, 2)	0.6	0.425	0.161	0.099	0.072	0.056	0.046
(4, 6, 2)	0.9	1.627	0.615	0.379	0.274	0.215	0.177
(4, 8, 3)	0.6	0.390	0.205	0.141	0.108	0.088	0.074
(4, 8, 3)	0.9	0.505	0.266	0.183	0.140	0.114	0.096

Per-stage leakages behind Corollary 4: at $t = 1$ the span is 2 or 3 against $|S| = 4$, with β between 0.51 and 0.71; at every $t \geq 2$ the span reaches $|S|$ and $\beta_t \leq 2.7 \times 10^{-15}$, giving a per-stage floor below 4.3×10^{-30} .